

Curriculum Vitae/Resume

Seunghun Oh

PhD Applicant

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👤 Google Scholar

Education

Seoul National University (SNU)

Mar 2020 - Aug 2026

BS in *Mechanical Engineering*, Double Major in *Computer Science*

- **GPA: 3.94/4.0, Major GPA: 3.97/4.0**

Top 3.8% (8th of 207, Mechanical Engineering, SNU)

- 21 months of military service included

Gyeonggi Science High School for the Gifted (GSHS)

Mar 2017 - Feb 2020

Major in *Physics and Mathematics*

- 1 Year Early Entrance

Research Interests

- Robot Perception: localization, SLAM, and 3D reconstruction for autonomous systems
- Multi-Sensor Fusion: LiDAR-camera-IMU integration for robust 3D understanding
- Autonomous Marine and Field Robotics: perception-driven navigation and reliable operation in complex environments

Publications

- **S. Oh**, J. Choe, D. Lee, D. Lee, S. Jeong, Y.-C. F. Wang, and J. Park*
“SVRecon: Sparse Voxel Rasterization for Surface Reconstruction” [arXiv:2511.17364](#)
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), under review, 2026.
- D. Lee, J. Lee, W. Kim, **S. Oh**, and J. Park*
“Articulated Furniture Recovery from Rest-State Multi-View Images”
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), under review, 2026.
- H. Song, D. Lee, **S. Oh**, M. Jung, and A. Kim*
“The City That Never Settles: Simulation-based LiDAR Dataset for Long-term Place Recognition under Extreme Structural Changes”
IEEE International Conference on Robotics and Automation (ICRA) Workshop, 2025.
Best Paper Award. [arXiv:2505.05076](#)
- **S. Oh**[†], Y. Kim[†], C. Song, and A. Kim*
“LiDAR Data Processing Algorithm for Robust 6-DoF Estimation Using Circular Patterns”
Journal of the Korean Robotics Society (KROS), June 2025. [KROS Journal Link](#)
- S. Hahn[†], **S. Oh**[†], M. Jung, A. Kim, and S. Jung*
“Quantitative 3D Map Accuracy Evaluation Hardware and Algorithm for LiDAR (-Inertial) SLAM”
International Conference on Control, Automation and Systems (ICCAS), 2024. [arXiv:2408.09727](#)

[†] Co-first authors. * Corresponding author.

Research Experiences

Undergraduate Researcher, SNU-VGI Lab (Advisor: Prof. Jaesik Park)

Seoul, Korea

Seoul National University

Sep 2025 – now

- **Developed a novel “Rest-State” framework for Articulated 3D Reconstruction**
Contributed ideas to the core pipeline design and ran baseline experiments on large-scale scenes.
- **Advanced SDF-based Sparse Voxel Rasterization with 3D Foundation Models**
Developed *SVRecon*, an SDF-based sparse voxel rasterization framework using Pi3-based initialization and hierarchical smoothness losses, achieving cleaner surfaces and faster convergence than SVRaster.

Undergraduate Researcher, RPM Robotics Lab (Advisor: Prof. Ayoung Kim)

Seoul, Korea

Seoul National University

Dec 2023 – Aug 2025

- **3D Reconstruction with SDF-based Sparse Voxel Rasterization**
Developed an SDF-based voxel rasterization pipeline to enhance mesh accuracy, extending SVRaster’s architecture. Focused on enforcing voxel-wise continuity using CUDA and PyTorch. → [Open-source project](#)
- **Long-term Place Recognition using Simulated LiDAR Datasets**
Adapted and implemented long-term loop closure algorithms (Scan Context, BTC, Solid) in simulated urban environments. Evaluated PR performance under structural changes with customized pipelines.

- **LiDAR-Camera Calibration via Circular Marker Detection**
Developed a robust LiDAR-based 6-DoF pose estimator using circular patterns, later integrated with *DiscoCal* (CVPR) into a full LiDAR-camera calibration system. → [Open-source project](#) 
- **Quantitative Evaluation of 3D SLAM Map Accuracy**
Designed experimental setups and metrics for evaluating LiDAR(-inertial) SLAM accuracy. Developed benchmarking tools for 3D map reconstruction quality comparison. → [Open-source project](#) 

Research Student, Aerospace Engineering Club (Bulnabi)
Seoul National University

Seoul, Korea
Dec 2023 – Apr 2024

- Built foundational robotics environment using Ubuntu and ROS
- Simulated drone operations with Gazebo and QGroundControl

Research Student, Mechanical Engineering Club (ZERO)
Seoul National University

Seoul, Korea
Mar 2020 – Aug 2020

- Focused on C/C++ development for low-level control modules in a Linux-based robotics setting.

Selected Awards and Honors

Scholarships / Fellowships

- **The National Presidential Science Scholarship**, Korea Student Aid Foundation 2024 – 2025
Full tuition plus approx. \$3,700/year stipend for **four semesters**, awarded by the Korean government to top STEM undergraduates.
- **Work-Study Scholarship** (*Value Exploration and Practice*), Seoul National University Mar 2025 – Nov 2025
Work-study research scholarship supporting an independent undergraduate project (approx. \$1,500).
- **Work-Study Scholarship** (Type 1), Seoul National University Mar 2024 – Nov 2024
Work-study research scholarship supporting an independent undergraduate project (approx. \$2,200).
- **Merit-based Scholarship**, Dept. of Mechanical Engineering, Seoul National University Spring – Fall 2021

Awards and Honors

- 2025 – Co-authored ICRA 2025 workshop paper [1]; awarded **Best Research Award** (1st Place), IEEE ICRA Workshop on Future of Construction
- 2025 – **Outstanding Undergraduate Research Opportunities Program Award**, Dept. of Mechanical Engineering, Seoul National University, selected **1 of 3** awardees from a **57-student** cohort (top ~5%)
- 2024 – **Grand Prize**, Mechatronics Competition
- 2022 – **Winner**, Startup Camp hosted by Hanbat University (During Military Service)
- 2021 – Academic Honor: Highest GPA in 2nd Semester (4.3/4.3)
- 2018 – **Training Lineup for IPHO (International Physics Olympiad)**
- 2018 – **Winner**, SNU Youth Engineering Frontier Camp

Presentations / Talks

- Poster Presentation, “LiDAR Data Processing Algorithm for Robust 6-DoF Estimation Using Circular Patterns”
Presented at KROS Spring Conference, Gangwon, Korea, 2025
- Seminar Talks, “Introduction to Linux”, “Drone software & simulation” “Introduction to ROS” [link](#) 
Robotics Club Seminar, Seoul National University, 2024

Leadership / Extracurricular Activities

Republic of Korea Air Force (ROKAF)
Sergeant, 15th Air Force Base, 256th Squadron

Seoul, Korea
Mar 2022 – Dec 2023

- Led a squad of 10 airmen, overseeing daily operations, safety, and training for unit readiness.
- Managed Confidential Level II documents and coordinated flight schedules with pilots.

Mentoring Volunteer Program
Volunteer Instructor and Team Leader

Andong, Korea
Jan 2025 – Feb 2025

- Served as volunteer instructor and mentor: taught physics classes, led Sports Day activities, and provided interview coaching and STEM mentoring for high school students in a one-week program.

Skills

C++, C, Java, Python, MATLAB, RISC-V Assembly, Bluespec, PyTorch, CUDA, ROS1/2, Arduino, Fusion 360, L^AT_EX